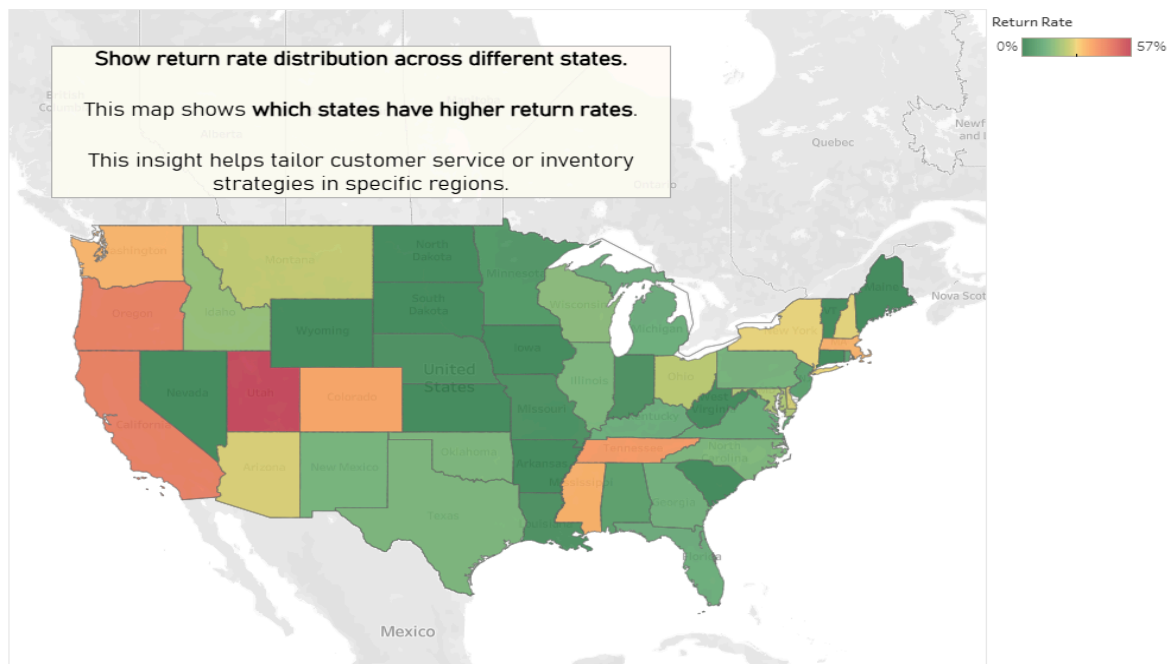


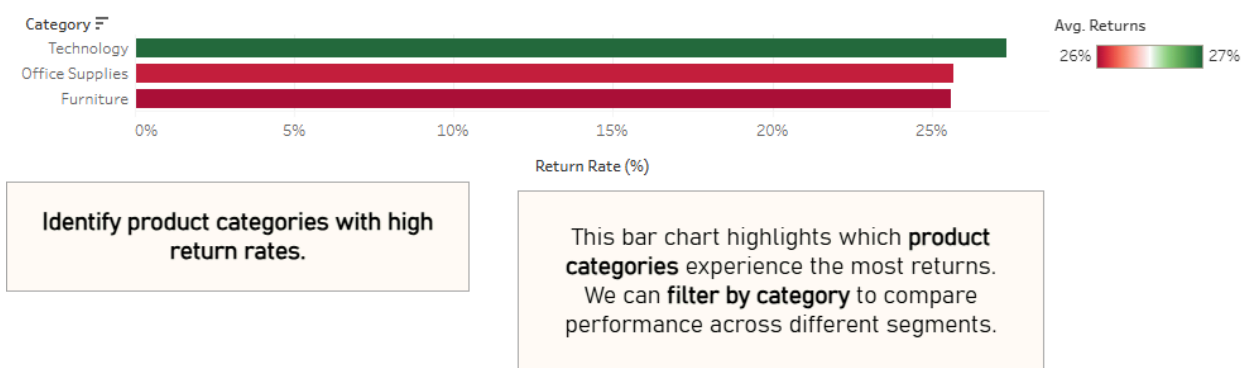
In this analysis, we aim to understand return rates by identifying key insights and potential actions to reduce returns. By examining return rates across different dimensions—geography, time, product categories, and customer behavior—we can develop data-driven strategies to improve business outcomes. Identifying patterns in average return rates allows us to investigate potential issues with specific products or assess the need for adjustments in certain product categories. The ultimate goal is to use these insights to minimize returns, optimize inventory management, and enhance profitability.

A critical question in this analysis is how returns should be measured. To effectively analyze return trends, we will primarily use return rate percentage, which is calculated as the total number of returned items divided by the total number of items sold. This metric provides a clear understanding of return trends relative to sales volume. In addition, we will also consider the total number of returns to assess the actual volume of products being returned, as well as total sales, which helps evaluate the financial impact of returns. A return rate of 10% may seem insignificant, but if the returned items are high-value products, the financial consequences could be substantial, leading to lower overall profitability.

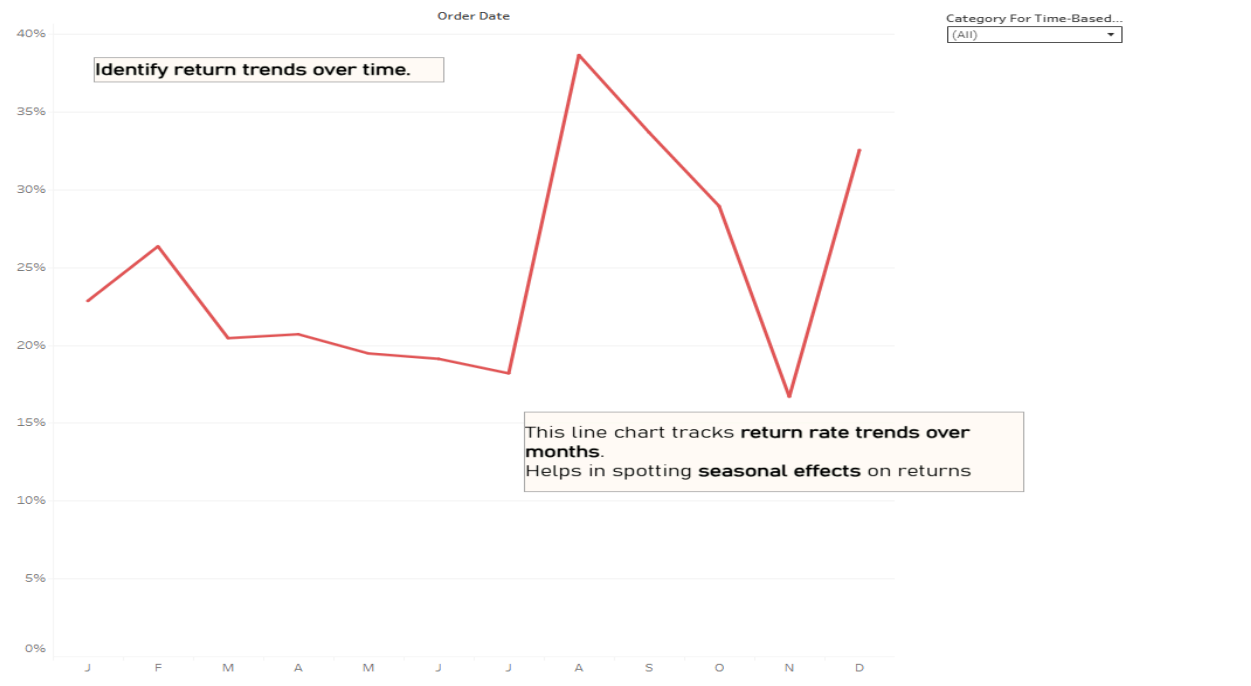
One of the key aspects of this analysis is understanding geographic return patterns. The return rate by geography is visualized through a color-coded map, where red and orange indicate higher return rates and green represents lower return rates. This visualization provides insights into regional return trends and helps businesses make informed decisions about inventory allocation and logistics strategies. For example, states such as Utah, Oregon, and California exhibit the highest return rates. Investigating these areas further may reveal issues related to shipping reliability, customer expectations, or product suitability in those regions.



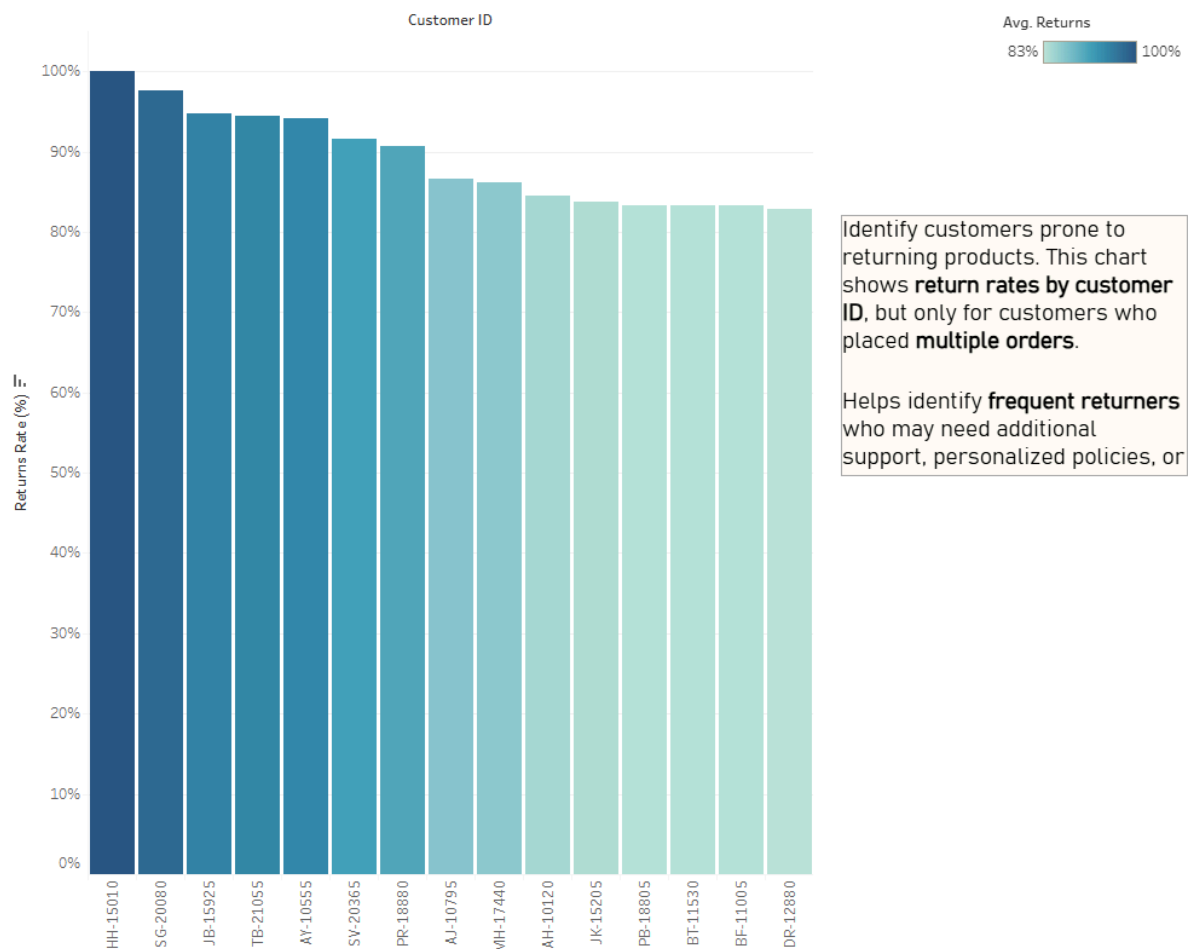
In addition to geography, product category is another critical factor influencing return rates. A bar chart displaying return rates by product category helps identify which types of products are most frequently returned. If a particular category, such as technology, shows an abnormally high return rate, it may indicate product defects, missing accessories, or customer dissatisfaction due to unclear product descriptions. Understanding these trends allows businesses to refine their return policies, improve product quality, and adjust marketing strategies to set more accurate customer expectations.



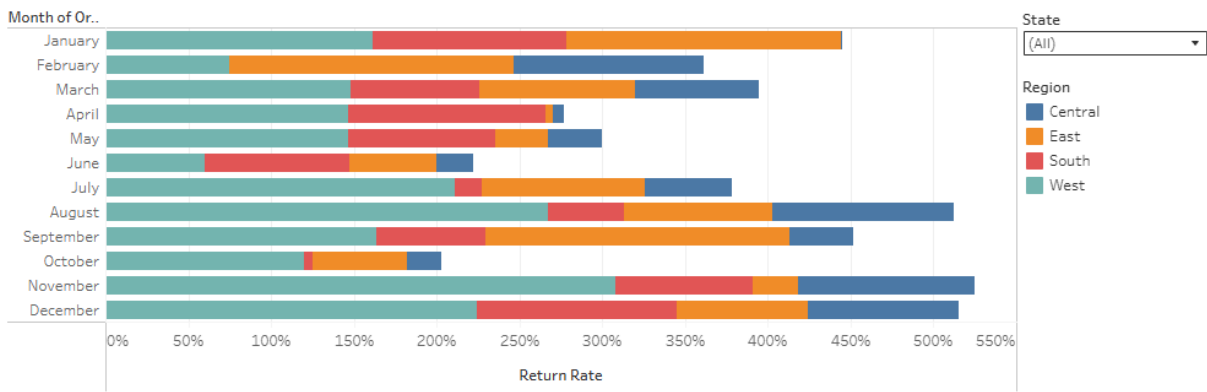
Examining return rates over time is also essential for recognizing seasonal return patterns. A time-based return rate analysis, visualized through a line chart, reveals fluctuations in return rates across different months. This insight helps businesses determine whether returns increase after major sales events, during specific seasons, or due to warranty-related claims. The analysis indicates that April has the highest return rate, suggesting a potential pattern influenced by factors such as post-holiday returns, warranty expirations, or shifting customer buying behaviors.



Customer behavior is another influential factor in return rates. A bar chart displaying the top 15 customers with the highest return rates helps highlight frequent returners. This visualization filters out one-time returners and focuses on customers with multiple orders, allowing businesses to identify patterns in repeat returns. If certain high-value or VIP customers consistently return products at a high rate, companies may consider personalized engagement strategies to address their concerns while maintaining their loyalty. Providing enhanced customer support, more detailed product information, or tailored return policies could help reduce return rates while improving customer satisfaction.

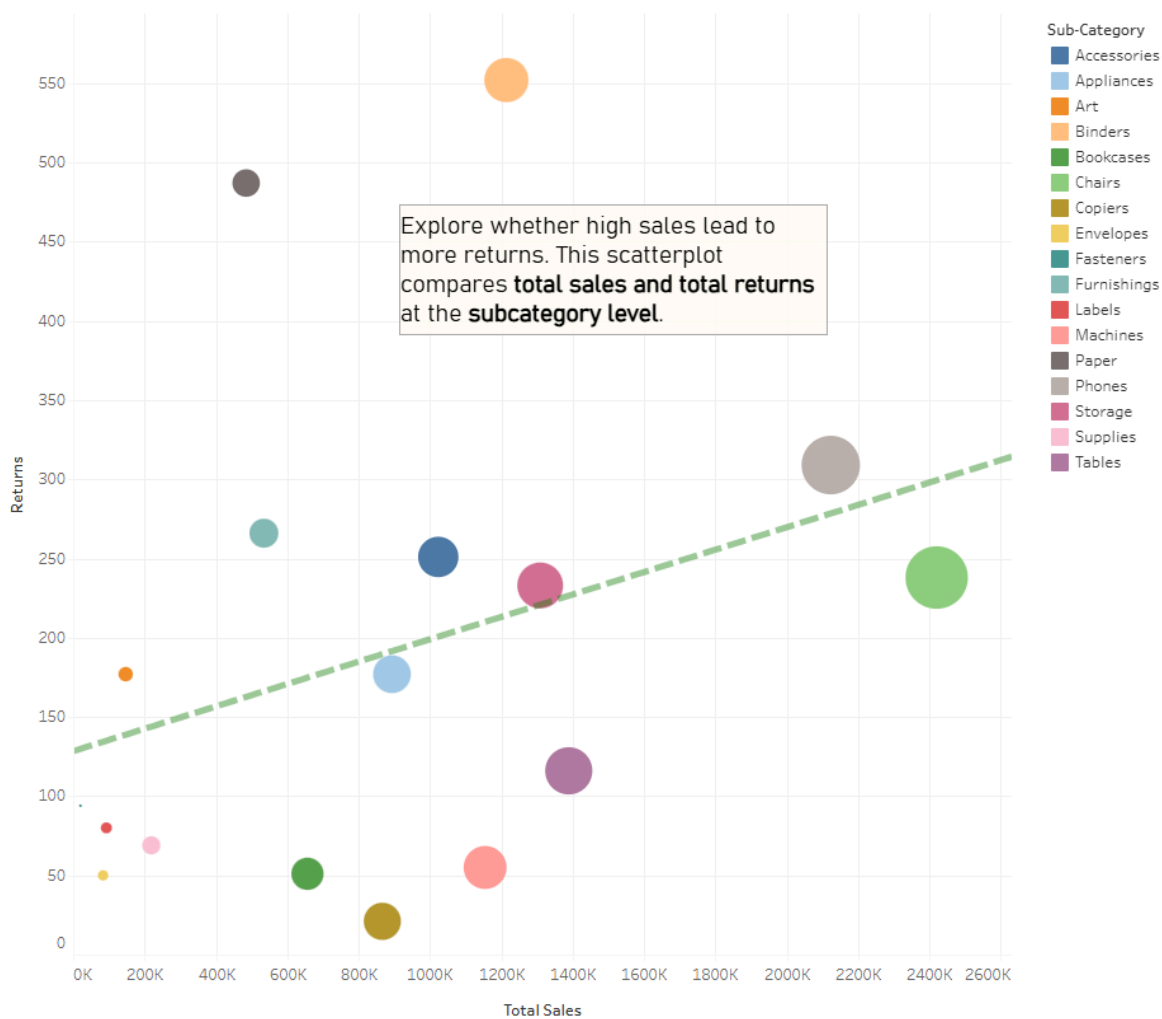


A more complex analysis involves return rates by both time and region, visualized through a stacked bar chart. This chart helps businesses explore whether return rates vary based on seasonal trends and geographic location. For instance, the data suggests that the Western region experiences higher return rates overall, which may indicate differences in customer expectations, logistical challenges, or regional product preferences. Understanding these insights can guide adjustments in shipping policies, customer support strategies, and marketing efforts tailored to specific regions.

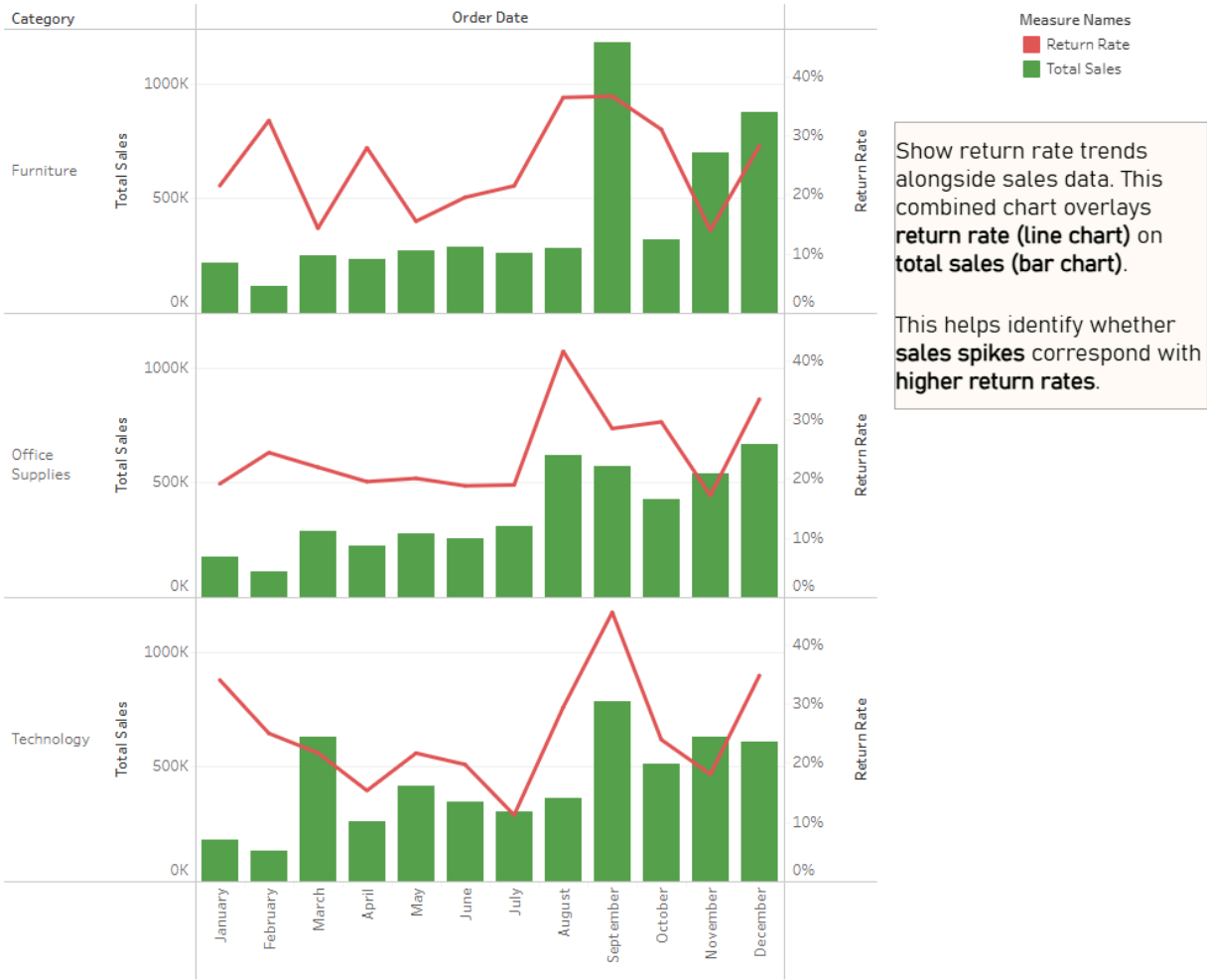


Compare return rates across different locations and time periods. We can see if specific regions experience **seasonal spikes in returns**. This helps businesses focus on **regional return policies** or **seasonal inventory adjustments**.

Another key relationship explored in this analysis is the correlation between sales and returns. A scatterplot visualization examines whether higher sales volumes lead to higher return rates. The trend line suggests a positive correlation, meaning that as sales increase, returns also tend to increase. This insight helps determine whether return rates are consistent across different products or if certain items experience disproportionately high returns. For example, if a high-selling smartphone model also has an extremely high return rate, it could indicate product defects, misleading advertisements, or unmet customer expectations. Identifying such trends allows businesses to take proactive measures, such as improving quality control, enhancing customer education, or refining marketing strategies.



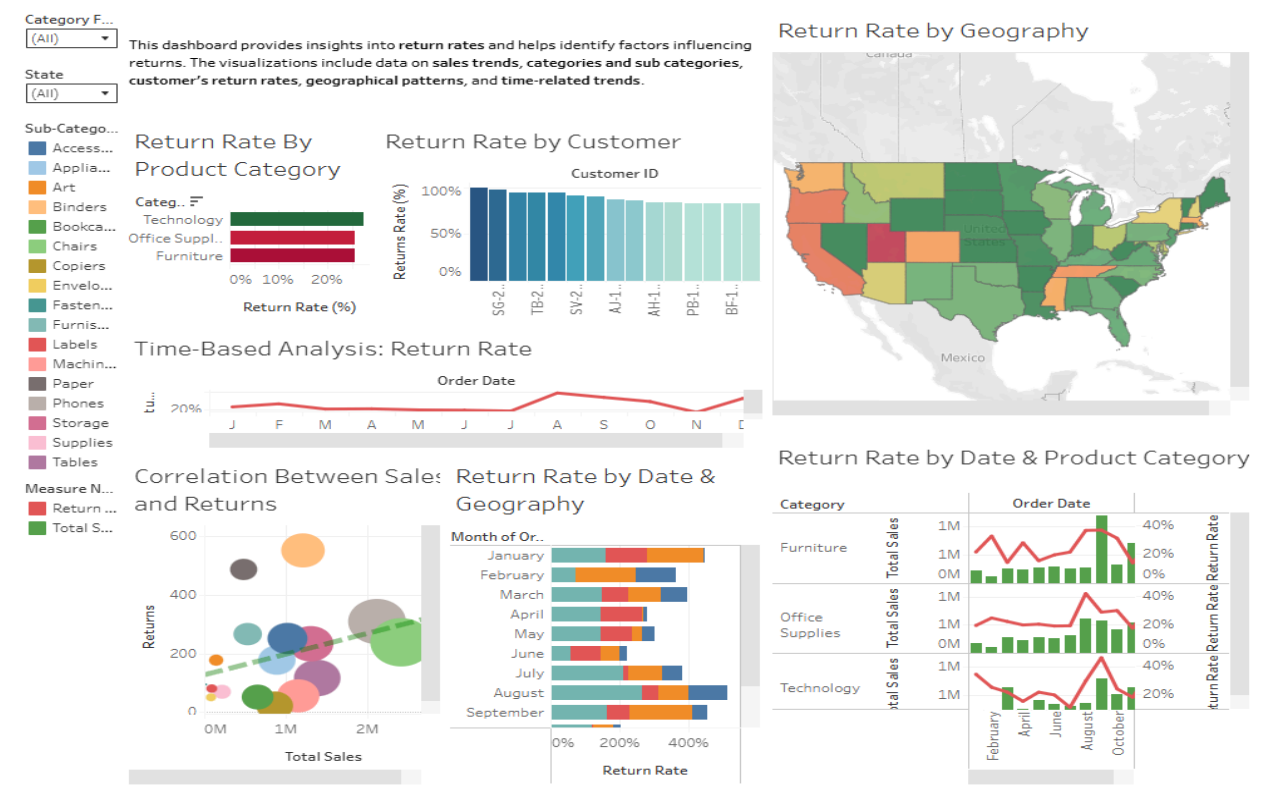
To provide a more comprehensive view, a return rate by date and product category analysis combines a bar chart for total sales with a line chart for return rates over time. This visualization helps determine whether sales spikes correlate with increased return rates. For example, major promotional events such as Black Friday or Cyber Monday often lead to both higher sales and higher return rates, as customers make impulsive purchases that they later regret. By understanding these trends, businesses can develop strategies such as enhancing product descriptions, offering better pre-purchase customer support, or implementing stricter return policies during peak sales periods.



Finally, the interactive dashboard provides four filters that allow users to dynamically analyze return trends:

- 1. Category Filter – Enables users to examine return rates across different product subcategories.
- 2. State Filter – Helps identify geographic return rate variations.
- 3. Date Filter – Allows users to analyze how return rates fluctuate over time.
- 4. Subcategory Selection – Provides deeper insights into specific product types.

By adjusting these filters, businesses can pinpoint specific causes of high return rates and develop targeted strategies to mitigate them.



In conclusion, based on the analysis presented, the primary region of concern is the Western region, where return rates are highest. Further investigation is needed to determine whether this trend is driven by shipping issues, customer service challenges, or product dissatisfaction.

In terms of product categories, furniture should be a key focus area for investment, as it generates the most profit and has the lowest return rate. Conversely, technology exhibits the

highest return rate, suggesting that funding and inventory allocation in this category should be carefully managed to minimize financial losses and improve customer satisfaction.